ing and mapping the sea floor using sonars and cameras.

A wide variety of sensor instruments are used in the ocean, including acoustic doppler current profiler, benthic flow meter, bottom pressure and tilt meter, conductivity-temperature-depth (CTD), dissolved oxygen sensor, digital still camera, high-definition video camera, hydrophone, mass spectrometer, optical attenuation sensor, pH and carbon-dioxide sensor, pressure sensor, remote access fluid and DNA sampler, resistivity probe, seismometer, sonar, thermistor array and turbulent-flow current meter.

Fig. 3: DO sensor (Credit: www.edaphic.com.au)

Ocean Sensor Systems manufactures sensors and wave instrumentation for use in oceans and seas, for the worst environmental conditions.

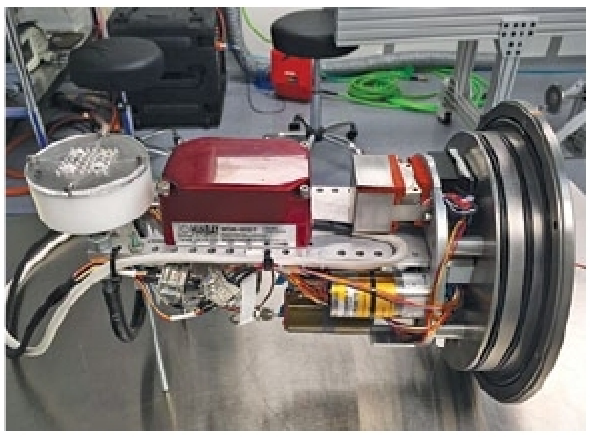

Fig. 4: Pressure sensor with quartz crystals (Credit: https://spectrum.ieee.org)

Fig. 5: SeaFET ocean pH sensor (Credit: www.oceanologyinternationalamericas.com)

Fig. 6: Multi-beam sonar (Credit: www.blueprintsubsea.com)

Sea-Bird Scientific manufactures various products for measuring salinity, fluorescence, nutrients and related oceanographic parameters in marine waters. These tools are used by ocean researchers and key industries engaged in offshore exploration and ocean resource utilisation.

There is no dearth of developments in the fields of deep-sea mining, research, communication, submarines and so on. This article covers some important sensors used for different deep-sea applications, including civil, military, research, communication, seismic and energy.

Types of deep-sea sensors

Let us explore the various types of deep-sea sensors.

DO sensor. Dissolved oxygen (DO) meter to assess water quality and measure the amount of dissolved oxygen in a liquid is one of the most important instruments because of its influence on the organisms living in water. DO sensor for depths of up to 6000 metres consists of a pre-amplifier covered by a titanium housing, as shown in Fig. 3. This complete sensor is used to interface CTD probe systems.

Pressure sensor. Seismic activity under the water is monitored by placing pressure sensors on the seafloor. Piezoelectric (quartz crystals) sensors produce electric charge when placed under pressure. These can measure pressure or weight of the water above. High or low pressure of different sections along a fault line determine where tectonic plates are locked up or are trying to move past each other. Pressure builds up between the plates, resulting in earthquakes when these break free.

Pressure sensors are prone to drift and lose accuracy over time. Electronics engineers from the University of Washington are now testing a low-cost, self-calibrating pressure sensor that could be deployed on the seafloor to monitor long-term seismic activity. Data from a deep-sea sensor network could help scientists understand what happens along fault lines.

pH sensor. Increasing atmospheric carbon-dioxide is driving long-term decrease in ocean pH. Measuring pH in sea water with glass electrodes demands a special sensor construction to avoid mistakes caused by high ionic strength of sea water. The ion sensitive field-effect transistor (ISFET) pH sensor based on Honeywell Durafet ISFET die can be immersed directly in seawater. It is capable of reporting pH with good accuracy.

Sonar. Remote underwater observation is much easier now, because sound navigation and ranging (sonar) technology has become much more advanced. Both passive and active sonar are used in modern naval warfare from water-borne vessels, aircraft and fixed installations. Submarines depend on sonar to a great extent for underwater communication. A new generation of multi-beam sonar designed for use across a wide variety of underwater applications is shown in Fig. 6.

Imaging sensor. Underwater photogrammetry in the deep sea is different from that of land or in space. Rough conditions, high pressures, absence of natural light and refraction are some problems. Attenuation and scattering degrade the radiometric image quality and limit effective visibility.

An optical system for AUVs for the purpose of visual mapping of large areas of the seafloor up to 6000 metres

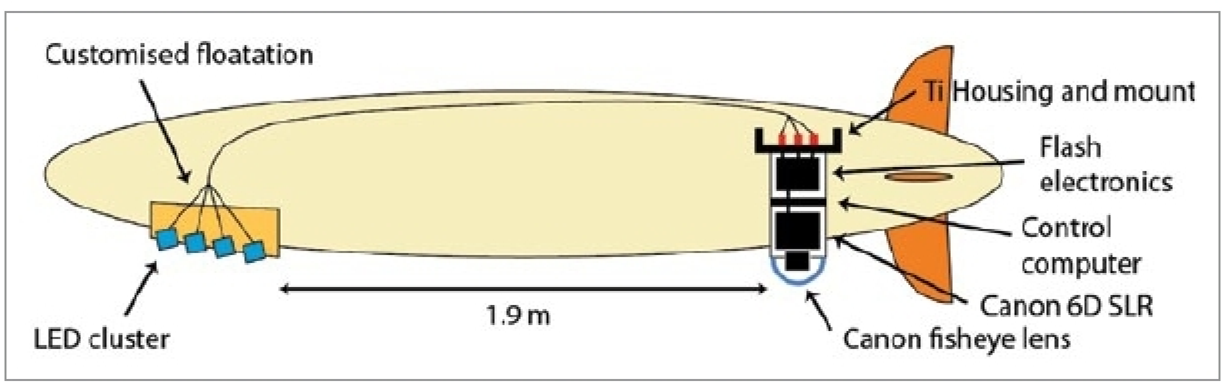

Fig. 7: Schematic overview of GEOMAR Remus 6000 AUV (Credit: www.mdpi.com)